measure. It could even point the way toward a theory uniting the forces, or help us to understand quantum gravity.

Having some kind of guidance from observations or experiments on how to improve on Concordance Cosmology or the Standard Model would be very helpful. Because over on the pure theory side of things, things are getting very, very weird.

THROUGH A GLASS DARKLY

I recently came across an old black-and-white photo of Paul Dirac, Nobel Laureate and pioneer of quantum mechanics, standing on the grounds of Princeton's Institute for Advanced Study with an axe slung over his shoulder. During his many visits there from the 1930s to the 1970s, he was known to wander through the woods behind the institute, clearing new paths for the resident theorists to walk and talk and think about the nature of reality. My own guide through those same muddy trails was Nima Arkani-Hamed, which seems appropriate, because he is a theorist determined to take an axe to our current understanding of quantum mechanics, and to the entire notion of spacetime itself.

Arkani-Hamed has been working on a way of calculating the interactions between particles using a completely new framework, one that starts from a kind of abstract mathematics in which space and time are not, strictly, included. The work is still in its early stages, and so far applies more to certain idealized systems than to experimental results. But if it works out, the implications could not be more mind-blowing. "What we're seeing is just, it's in baby, baby toy, toy, toy examples, right? You can justifiably use as many diminutives as you want on what's actually been accomplished, and I would be entirely sympathetic," he tells me. "But for what it's worth, there is starting to be one or two examples of actual concrete physical systems not so far from what we see in the real world where we can actually figure out how to describe them without either spacetime or quantum mechanics." I tell him I'm trying to wrap my head around what it means to live in a universe where space and time aren't real. He laughs. "Join the club."

Before you dismiss the idea as eccentric-theorist hyperbole, I should point out that Arkani-Hamed is not the only one talking like this. "I'm sure you've heard this from many people," Clifford Johnson tells me,